COMSCI / ECON 206 | PS2 RESEARCH SYMPOSIUM | FALL 2026

AI, Rent-Seeking, and Ticket Allocation: Mechanism Design under Asymmetric Participation Costs

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10 REDUCED INEQUALITIES

FINAL A0 POSTER

KEY MESSAGE — **AI changes the cost of strategic participation. In our stylized model, effort contests are fragile to asymmetric cost reductions; verified lotteries remove speed advantage but weaken when extra identities become cheap.**

01 Question & model

Research question & verified gap

RQ: How does AI's asymmetric reduction of ticket-buying costs change strategic competition and welfare, and which allocation mechanism is more robust?
Setting: 200 genuine fans, 50 scalpers, one representative scarce ticket; official price $100, resale price $180.
Verified gap: resale, ticket auctions, and speed races are studied separately; our comparison treats AI as an asymmetric participation-cost shock across two allocation rules.

Strategic model & foundations

Players: fans value attendance; scalpers value resale margin.
Actions: effort e_i in a Tullock contest; identities q in a verified lottery.
Payoff: $V_i \cdot e_i/\Sigma e_j - c_i e_i$. Platform commits to the rule first.
Solution concept: exact heterogeneous single-prize Nash equilibrium.
AI stress test: $(c_F, c_S)=(10,10) \rightarrow (5,1)$; identity cost $k=0.25 \rightarrow 0.10$.

Three complementary lenses

Game theory — lower marginal cost raises equilibrium effort; asymmetric reductions can change the active set.
Social choice — compare primary access, final ownership, consumer surplus, scalper profit, and rent dissipation.
Mechanism design — test effort contest versus verified lottery; randomization removes speed advantage but not cheap multi-identity manipulation.

Auction or allocation comparison

Scarce resource: one representative ticket.
Effort Contest: allocation probability $e_i/\Sigma e_j$; all effort costs are paid; winner pays $100.
Verified Lottery: one entry per identity; extra identities cost k; winner pays $100.
Stopping rule: one ticket allocated, then possible resale at $180.
Boundary: synthetic single-ticket comparative statics, not a full concert inventory or real-world causal estimate.

02 Design & evidence



Figure 1. Evidence chain: cost shock → strategic response → mechanism comparison → observed synthetic outcomes → design revision.

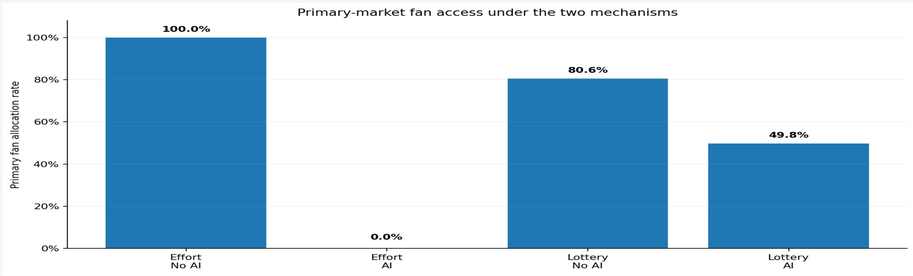


Figure 2. Under the stated AI stress test, primary fan allocation changes 100.0%→0.0% in the Effort Contest and 80.6%→49.8% in the Verified Lottery. Sensitivity reveals a nonlinear effort threshold between $c_S=3$ and 2; when identity cost falls to 0.05, scalpers use 7 identities and fan allocation falls to 36.2%. Final fan ownership remains 100% in all four baseline conditions, so primary access and final ownership must be evaluated separately.
Verification: seed 2026 · 5,000 simulations/condition · 500/sensitivity value · runtime ≈10.9 s · GitHub commit ce6cb2b. Evidence status: observed synthetic output.

03 Impact & dialogue

Practical impact: a real-world case

Case: high-demand concert ticketing platforms.
Decision: should access depend on speed/effort or verified randomization?
Potential benefit: reduce scalable automation advantage in the primary market.
Risk: verification can exclude legitimate users; cheap identity creation can undermine lotteries.
Before adoption: measure duplicate-entry rates, verification failures, resale flows, and primary allocation before/after rule changes.
Correction route: sensitivity analysis, peer challenge, and validation against platform data.

Interdisciplinary evidence

Behavioral science · Hugging Face
Users choose a role, predict effort and mechanism preference, then see the benchmark and reflect. Classroom evidence is exploratory, not population validation.

Computer science · GitHub / Colab
Exact heterogeneous Tullock benchmark + Monte Carlo comparison; fixed seed, parameter sweeps, saved outputs, runnable notebook.

Integration
Behavioral predictions probe the rational cost-response assumption; computational sensitivity shows where mechanism robustness breaks.

Limitations & next test

Main limitation: AI cost reductions are stipulated, not estimated; one-ticket abstraction omits full inventory dynamics and heterogeneous resale frictions.
Next test: estimate realistic effort/identity costs, collect structured peer play, and compare a price-based auction as a third mechanism.
Symposium question: Which real-world verification cost or behavioral response would most likely overturn our mechanism ranking?

SEP 28 Symposium, peer evaluation & case impact
SEP 30 Final update

Acknowledgments

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Selected references

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Leslie, P. & Sorensen, A. (2014). Resale and Rent-Seeking. REStud 81(1), 266–300. doi:10.1093/restud/rdt033
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OPEN MATERIALS

Hugging Face: https://huggingface.co/spaces/dku-comsci-econ206-2026/ai-ticket-allocation
GitHub: https://github.com/dku-comsci-econ206-Autumn2026/PS2-Adaptive-Ticket-Auctions · commit ce6cb2b
Colab: https://colab.research.google.com/drive/101hr5eAFEfbxkwLoj-67jMKfCMvo6iC_